

Chiller Plant Retrofit Programme - Business Case & Financial Review

Keppel Bay Tower · Office · Singapore

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1. Investment summary

This paper reviews the financial performance of the Keppel Bay Tower chiller plant optimisation retrofit completed in October 2024 and recommends a phased rollout to other office assets.

Capex item	S\$
AI optimisation platform and integration	380,000
Variable speed drives (pumps and cooling tower fans)	450,000
M&V instrumentation	210,000
Controls and AHU upgrades	520,000
Commissioning and contingency	340,000
Total capex	1,900,000
Less green building grant (assumed)	(250,000)
Net capex	1,650,000

2. Business-case assumptions (approved 2023)

- Planned electricity saving of 1.76 GWh per year at an average tariff of S\$0.29/kWh, plus S\$16,000 per year of avoided maintenance: S\$526,000 per year in total.
- Planned simple payback: 3.6 years gross, 3.1 years after the grant.

3. Actual performance (first 12 months)

Item	Business case	Actual
Electricity saved (GWh)	1.76	1.76
Average tariff (S\$/kWh)	0.29	0.30
Energy cost saving (S\$)	510,000	528,000
Maintenance saving (S\$)	16,000	20,000
Total annual saving (S\$)	526,000	548,000
Simple payback, net of grant (years)	3.1	3.0
NPV at 8% over 10 years (S\$)	1.88M	2.03M
IRR	29%	31%

4. Value impact

The annual saving flows directly to net property income (NPI). At an indicative capitalisation rate of 4.0%, an NPI uplift of S\$548,000 corresponds to roughly S\$13.7 million of asset value, more than seven times the capital invested. The retrofit also helped the asset qualify for a sustainability-linked loan margin reduction of 5 basis points.

5. Risks observed

- Tariff volatility moves the payback in both directions; savings were measured in kWh first and valued second.
- Measurement and verification disputes were avoided because permanent M&V instrumentation was installed before the retrofit.
- Tenant disruption was minimal thanks to phased commissioning.

6. Portfolio rollout recommendation

Asset	Plant efficiency (kW/RT)	Est. capex (\$)	Est. saving (\$/yr)	Payback (yrs)
Tanjong Quay Tower	0.92	1,400,000	450,000	3.1
Southbank Office Tower	0.88	1,250,000	330,000	3.8

7. Recommendation

Approve phase 2 for Tanjong Quay Tower in FY2026 and Southbank Office Tower in FY2027, reusing the same vendor framework, M&V specification and shadow-mode commissioning. Cooling-intensive assets elsewhere in the wider portfolio should be screened using the same method, because savings scale with cooling load and operating hours.